Quizzes and Answer Keys

Short quizzes for every session, with a full answer key for teachers

HOW TO USE THIS DOCUMENT

- **Part A** (questions) can be printed on its own and handed to students. **Part B** (answers) is for teachers: print or keep it separately.
- Use quizzes as an *exit ticket* at the end of a session, as a *starter* the next day, or as a team game.
- Ask students to **explain why** the wrong options are wrong — that shows real understanding.

Part A — Questions

Day 1 — From DNA to Protein

#	Multiple choice
1	Which part of an amino acid is different in each of the 20 amino acids? A) The amino group B) The carboxyl group C) The R-group (side chain) D) The central hydrogen
2	How many essential amino acids must adults get from food? A) 5 B) 9 C) 11 D) 20
3	A nucleoside is made of... A) a base only B) a sugar and a phosphate C) a sugar and a base D) a sugar, a base and a phosphate
4	Which base pair has three hydrogen bonds? A) A–T B) G–C C) A–U D) A–G
5	Which enzyme unwinds the DNA double helix at a replication fork? A) DNA ligase B) Primase C) Helicase D) RNA polymerase

#	Multiple choice
6	Short pieces of DNA made on the lagging strand are called... A) Okazaki fragments B) Exons C) Primers D) Codons
7	The usual start codon codes for... A) Stop B) Methionine C) Alanine D) Glycine
8	Which ribosome site receives the incoming charged tRNA? A) A site B) P site C) E site D) S site

#	Short answer
9	Why is adenosine a nucleoside and not a nucleotide?
10	A DNA template strand reads 3′-TACGGA-5′. Write the mRNA and the amino acids.
11	Why does DNA replication need an RNA primer?
12	Why does an imprecise removal of an intron cause a faulty protein?
13	Name the three tRNA sites on the ribosome and give the job of each.
14	Write the complementary DNA strand for 5′-GAATTCCAAA-3′.

Day 2 — Lac Operon and Quorum Sensing

#	Multiple choice
1	An operon is... A) one gene with one promoter B) a group of genes controlled by one promoter C) a type of RNA D) a bacterial toxin
2	When lactose is absent, the lac repressor... A) binds to the operator B) binds to allolactose C) leaves the DNA D) is destroyed
3	Which molecule makes the lac repressor let go of the operator? A) Glucose B) cAMP C) Allolactose D) X-gal

#	Multiple choice
4	With lactose and glucose both present, lac operon expression is... A) high B) low C) zero D) unaffected by CAP
5	CAP–cAMP switches the operon... A) off when glucose is low B) on strongly when glucose is low C) on when lactose is absent D) off when lactose is present
6	Quorum sensing allows bacteria to... A) sense their population density B) digest lactose C) copy their DNA D) swim faster

#	Short answer
7	Why does *E. coli* ignore lactose when glucose is plentiful?
8	A mutation stops the repressor from binding the operator. What is the effect?
9	Why is it useful for a pathogen to use quorum sensing?

Day 3 — Plasmid Design and Cloning

#	Multiple choice
1	Which is NOT a minimum part of a lab plasmid? A) Origin of replication B) Selectable marker C) Cloning site D) Nucleus
2	EcoRI recognises... A) GAATTC B) CCCGGG C) AAGCTT D) GGATCC
3	Which enzyme seals the nicks between insert and plasmid? A) Helicase B) Taq polymerase C) DNA ligase D) EcoRI

#	Multiple choice
4	Which transformation step makes DNA enter the cell? A) Heat shock at 42 °C B) Boiling C) Freezing at −20 °C D) Adding X-gal
5	A colony on ampicillin + X-gal + IPTG is blue. It contains... A) no plasmid B) an empty plasmid C) a recombinant plasmid D) a dead cell

#	Short answer
6	Why do we cut the insert and the plasmid with the same restriction enzyme?
7	What does ligase do that hydrogen bonding between sticky ends cannot?
8	What is the purpose of the no-DNA control?
9	Why can a white colony still have the wrong plasmid?
10	Calculate: 0.5 ng of plasmid gives 150 colonies. What is the transformation efficiency?

Bacterial culture and agar plates

#	Multiple choice
1	An organism that cannot grow in the presence of oxygen is a... A) obligate aerobe B) obligate anaerobe C) facultative anaerobe D) microaerophile
2	Which phase of bacterial growth has the fastest, steady division? A) Lag B) Log (exponential) C) Stationary D) Death
3	MacConkey agar is mainly... A) an enriched medium B) a selective and differential medium C) a liquid medium D) a medium for fungi only
4	Why are plates incubated upside down? A) To save space B) So that water drops do not fall on the colonies C) To keep out oxygen D) To make agar set

#	Short answer
5	Why is agar a good solidifying agent for culture media?
6	What is the difference between selective and differential media?
7	Starting with 1 cell and a generation time of 30 minutes, how many cells after 3 hours?
8	Why do we re-sterilise the loop between quadrants when streaking?
9	Why should plates from hand prints never be opened?

Viral diagnosis and PCR

#	Multiple choice
1	Why is Taq polymerase used in PCR? A) It works at low temperature B) It is heat-stable C) It makes RNA D) It cuts DNA
2	Primers are... A) long double-stranded DNA B) short single-stranded DNA that start copying C) enzymes D) fluorescent dyes
3	How many copies after 25 perfect cycles from one molecule? A) 25 B) 625 C) about 33 million D) about 1 billion
4	A lower Ct value means... A) less virus B) more virus C) a failed test D) a different virus
5	In gel electrophoresis, DNA moves toward the... A) negative electrode B) positive electrode C) top of the gel D) well

#	Short answer
6	Why is reverse transcriptase needed to detect SARS-CoV-2?
7	Rank patients a (Ct 23), b (30), c (18), d (20), e (37) by viral load, highest first.
8	Which patient is most severely ill?
9	Why do real PCR experiments include positive and negative controls?

Enzymes

#	Multiple choice
1	An enzyme speeds up a reaction by... A) adding energy B) lowering the activation energy C) changing the amount of product D) being used up
2	Boiling an enzyme usually... A) speeds it up B) denatures it C) has no effect D) makes more substrate
3	Pepsin works best at... A) pH 2 B) pH 7 C) pH 9 D) pH 12

#	Short answer
4	Why does an enzyme not change the final amount of product?
5	Why do we need vitamins in the diet, in terms of enzymes?
6	Which cup foamed the most in the potato experiment and why?

Ecology and biomagnification

#	Multiple choice
1	Roughly what fraction of energy passes to the next trophic level? A) 90 % B) 50 % C) 10 % D) 1 %
2	Biomagnification is... A) a poison building up in one animal over its life B) a poison becoming more concentrated up the food chain C) a toxin breaking down D) a plant growing bigger
3	Which is a microbial biopesticide? A) DDT B) Neem leaf C) Bacillus thuringiensis D) Sex pheromone

#	Short answer
4	Why do food chains rarely have more than five levels?
5	Why is DDT more dangerous for an osprey than for a grasshopper?
6	Give one advantage and one limitation of biopesticides.
7	Order these from producer to apex predator: tiger, grass, chital deer.
8	In the sweets model, what are the expected DDT amounts per grasshopper, frog and top predator?

Part B — Teacher answer key

Day 1 — From DNA to Protein

#	Answer	Why
1	C	The amino group, carboxyl group and hydrogen are the same in all amino acids; only the R-group changes.
2	B	Histidine, isoleucine, leucine, lysine, methionine, phenylalanine, threonine, tryptophan and valine.
3	C	Adding a phosphate group to a nucleoside makes a nucleotide.
4	B	G–C has three; A–T (and A–U) has two.
5	C	Helicase breaks the hydrogen bonds between the strands.
6	A	The lagging strand is made discontinuously, as Okazaki fragments.
7	B	AUG = methionine = start.
8	A	A = arrival. P holds the growing peptide; E is the exit.

#	Model answer
9	It has a sugar (ribose) and a base (adenine) but no phosphate group.
10	mRNA: 5′-AUGCCU-3′. Amino acids: Met – Pro.
11	DNA polymerase can only add nucleotides to an existing 3′ end; it cannot start a new strand from nothing. Primase makes a short RNA primer to give it a start.
12	Codons are read in threes. Extra or missing bases shift the reading frame (frameshift), so every codon after the mistake is read wrongly.
13	A (arrival): accepts the new charged tRNA. P (peptide): holds the tRNA with the growing chain. E (exit): the empty tRNA leaves from here.
14	3′-CTTAAGGTTT-5′.

Day 2 — Lac Operon and Quorum Sensing

#	Answer	Why
1	B	Genes in an operon are transcribed together as one mRNA.
2	A	Bound to the operator, it blocks RNA polymerase.
3	C	Allolactose is made from lactose and binds the repressor.
4	B	Repressor is inactive, but CAP is not helping because glucose is high.
5	B	Low glucose gives high cAMP, which binds CAP and helps RNA polymerase.
6	A	Signal molecules build up when the population is dense.

#	Model answer
7	Glucose gives energy faster and more easily. Making the lac enzymes while glucose is available would waste energy and materials.
8	The operon is on all the time, even without lactose (constitutive expression).
9	It can wait until there are enough cells for the attack to succeed, and so avoid alerting the immune system too early.

Day 3 — Plasmid Design and Cloning

#	Answer	Why
1	D	Bacteria have no nucleus; plasmids need ori, a marker and a cloning site.
2	A	EcoRI: G^AATTC.
3	C	Ligase forms the phosphodiester bonds.
4	A	Calcium-treated, cold cells take up DNA after a brief heat shock.
5	B	Intact lacZ makes β-galactosidase, which turns X-gal blue.

#	Model answer
6	So both have matching (complementary) sticky ends that can base-pair.
7	It forms strong covalent phosphodiester bonds and seals the backbone, so the join is permanent.
8	It should give no colonies on antibiotic plates. If colonies grow, the cells, plates or antibiotic are contaminated or faulty.
9	White only shows the lacZ gene was interrupted. The insert could be the wrong size or fragment, or the plasmid may have changed. It must be checked by PCR, digest or sequencing.
10	0.5 ng = 0.0005 µg. 150 ÷ 0.0005 = 300,000 = 3×10^5 colonies per µg.

Bacterial culture and agar plates

#	Answer	Why
1	B	Obligate anaerobes are harmed by oxygen.
2	B	Cells double steadily in log phase.
3	B	It selects Gram-negatives and shows lactose fermenters as pink colonies.
4	B	Condensation collects on the lid and would run over the agar.

#	Model answer
5	It sets firmly, melts only near boiling, stays solid at incubator temperatures, and very few bacteria can digest it.

#	Model answer
6	Selective media stop unwanted organisms growing; differential media let different organisms grow but look different.
7	3 hours = 6 generations. 2^6 = 64 cells.
8	To kill carried-over cells, so that each quadrant has fewer cells and gives single colonies.
9	They may contain unknown bacteria, some of which can cause infection. Plates stay sealed and are disinfected before disposal.

Viral diagnosis and PCR

#	Answer	Why
1	B	It survives the repeated heating to about 95 °C.
2	B	About 18–25 bases that bind either side of the target.
3	C	2^{25} = 33,554,432.
4	B	The signal crossed the threshold earlier because there was more starting material.
5	B	DNA is negatively charged.

#	Model answer
6	The virus has an RNA genome, and Taq polymerase copies only DNA. Reverse transcriptase first makes a DNA copy (cDNA).
7	c (18) > d (20) > a (23) > b (30) > e (37). Patient e is effectively negative or non-diagnostic.
8	Cannot be told from Ct alone. Patient c has the highest viral load, which suggests the most virus and perhaps the most infectious, but severity also depends on age, immunity, other illnesses and symptoms. A Ct value is not a measure of severity.
9	The positive control shows the reaction works; the negative control shows there was no contamination. Without them the result cannot be trusted.

Enzymes

#	Answer	Why
1	B	It provides an easier path for the reaction.
2	B	The protein unfolds and the active site loses its shape.
3	A	It works in the acidic stomach.

#	Model answer
4	It speeds up the forward and backward reactions alike, so the final balance (equilibrium) stays the same; it only reaches it faster.

#	Model answer
5	Many coenzymes are made from vitamins, and many enzymes cannot work without their coenzymes.
6	Raw potato at room temperature: its catalase is active. Boiled potato has denatured catalase; cold potato has slower catalase.

Ecology and biomagnification

#	Answer	Why
1	C	The rest is used or lost as heat.
2	B	Bioaccumulation is the build-up in one organism.
3	C	Bt is a bacterium; neem and pheromones are biochemical.

#	Model answer
4	Only about 10 % of energy is passed on at each level, so very little is left for a sixth level.
5	DDT is persistent and stored in fat. The osprey eats many contaminated fish, each with DDT from many smaller organisms, so it accumulates far more.
6	Advantage: usually less toxic and more specific. Limitation: farmers need knowledge to use them properly, and they may work more slowly or need more careful timing.
7	Grass → chital deer → tiger.
8	About 1, 3 and 9 units (they rise about threefold at each step).